A minor project on

"ADVANCED EVM [ELECTRONIC VOTING MACHINE]"

A project report submitted in partial fulfillment of the requirements for the 6th semester of the degree of

BACHELOR OF TECHNOLOGY
in
Electronics Engineering(IOT)

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DECLARATION

I hereby certify that the work which is being presented in this project report entitled **"ADVANCED EVM [ELECTRONIC VOTING MACHINE]"**, in partial fulfillment of requirements for the 6th semester of degree of BACHELOR OF TECHNOLOGY in ELECTRONICS ENGINEERING, submitted to the Department of Electronics Engineering, Faculty of Engineering and Technology, J.C. Bose University of Science and Technology, YMCA, Faridabad, Haryana-121006 is an authentic record of my work carried out during period from Month, year to Month, year, under the supervision of **Mr. Mukesh**. The matter presented in this project report, I has not submitted to any other University/Institute for the award of B.Tech. or any degree or diploma.

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CERTIFICATE

This is to certify that the project entitled **"ADVANCED EVM [ELECTRONIC VOTING MACHINE]"** submitted to Department of Electronics Engineering, Faculty of Engineering and Technology, J. C. Bose University of Science and Technology, YMCA, Faridabad, Haryana-121006 by **Mr. Mukesh** in partial fulfillment of the requirement for the 6th semester of degree of BACHELOR OF TECHNOLOGY in ELECTRONICS ENGINEERING, is a bonafide work carried out by him/her under my supervision and guidance. This project work comprises of original work and has not been submitted anywhere else for any other degree to the best of my knowledge.

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ABSTRACT

In modern democracies, ensuring the integrity, security, and transparency of the electoral process is paramount. Traditional voting methods often face challenges such as identity theft, proxy voting, and delayed result compilation. This project introduces an Advanced Electronic Voting Machine (EVM)—a secure, automated, and tamper-proof voting prototype designed to eliminate fraudulent voting and streamline voter authentication using an embedded microcontroller architecture.

The core system integrates biometric technology with real-time audio-visual feedback to create a seamless user experience. To enforce the "one person, one vote" rule, an optical fingerprint sensor is utilized for biometric authentication. Voters must scan their fingerprint against a pre-registered database; the system unlocks the voting interface only upon successful verification, effectively preventing duplicate voting.

The user interface consists of tactile push-buttons for casting votes, paired with pull-up/pull-down resistors to ensure clean, noise-free digital signals. Visual guidance is provided by a 16x2 LCD display, which utilizes an I2C module to drastically reduce wiring complexity and optimize microcontroller pin utilization. The LCD guides the voter through the authentication and voting phases step-by-step. To ensure transparency and acknowledge successful data entry, a buzzer is integrated into the circuit, providing a distinct audible beep confirming that a vote has been officially registered and recorded.

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PROBLEM STATEMENT

Critical challenges that affect the fairness, security, and efficiency of the voting process:

- **Duplicate voting**, This can occur due to weak verification mechanisms or lack of proper identity validation.
- **Impersonation**, where an unauthorized person casts a vote on behalf of someone else .
- **Lack of strong authentication**, Conventional systems rely on voter ID cards or manual verification, which can be forged, lost, or misused.
- **Manual counting of votes**, which increases the chances of human error and delays in result declaration.
- **lack of transparency** and real-time monitoring in existing systems.

To address these challenges, **there is a need for a modern voting system** that ensures:

- **Accurate and secure** voter authentication.
- **Prevention of duplicate and fake voting.**
- **Fast and automated vote recording and counting**.
- **Real-time result display**.
- Improved **transparency and reliability.**

INTRODUCTION

Voting is one of the most important processes in a democratic system, as it allows citizens to choose their representatives fairly and freely. However, traditional voting methods such as paper ballots and ordinary Electronic Voting Machines (EVMs) suffer from several limitations including duplicate voting, fake identity usage, manual errors, longer processing time, and lack of transparency. These issues can reduce the reliability and efficiency of the election process.
To overcome these problems, this project presents an Advanced Electronic Voting Machine using Fingerprint Authentication and IoT technology. The main objective of this system is to provide a secure, transparent, and efficient voting process by ensuring that only authorized voters are allowed to cast their votes. In this system, a fingerprint sensor is used for voter verification, which prevents unauthorized access and eliminates the possibility of multiple voting by the same person. The proposed system is developed using the ESP32 microcontroller, which provides both processing capability and internet connectivity. The voting process is simple and user-friendly. Once the voter's fingerprint is successfully verified, the voter can select the desired candidate using push buttons. The complete process and instructions are displayed on an LCD screen for better user interaction. After voting, the data is securely updated and can also be transmitted to an online platform in real time through IoT connectivity, making result monitoring faster and more reliable.

REQUIRED COMPONENTS

Hardware Required:

- ESP 32 Microcontroller
- Fingerprint Sensor (R307)
- LCD Display
- I2C Module
- Resistors
- Push Buttons (auto reset)
- Buzzer
- Buck Conveter

Software Required:

- Arduino IDE
- Embedded C
- IoT Platform (ThingSpeak / Firebase)

ESP32 MICROCONTROLLER

ESP32 is the central controller of the system with built-in Wi-Fi and multiple GPIOs. Initially, it is programmed via Arduino IDE and configured to communicate with all peripherals. It reads data from the fingerprint sensor and push buttons through serial and digital pins. It also drives the LCD using I2C communication. In the project, ESP32 performs authentication logic, vote counting, and data handling. It sends voting data to an IoT platform over Wi-Fi in real time. It ensures synchronization between input, processing, and output units. Overall, it acts as the brain of the voting machine

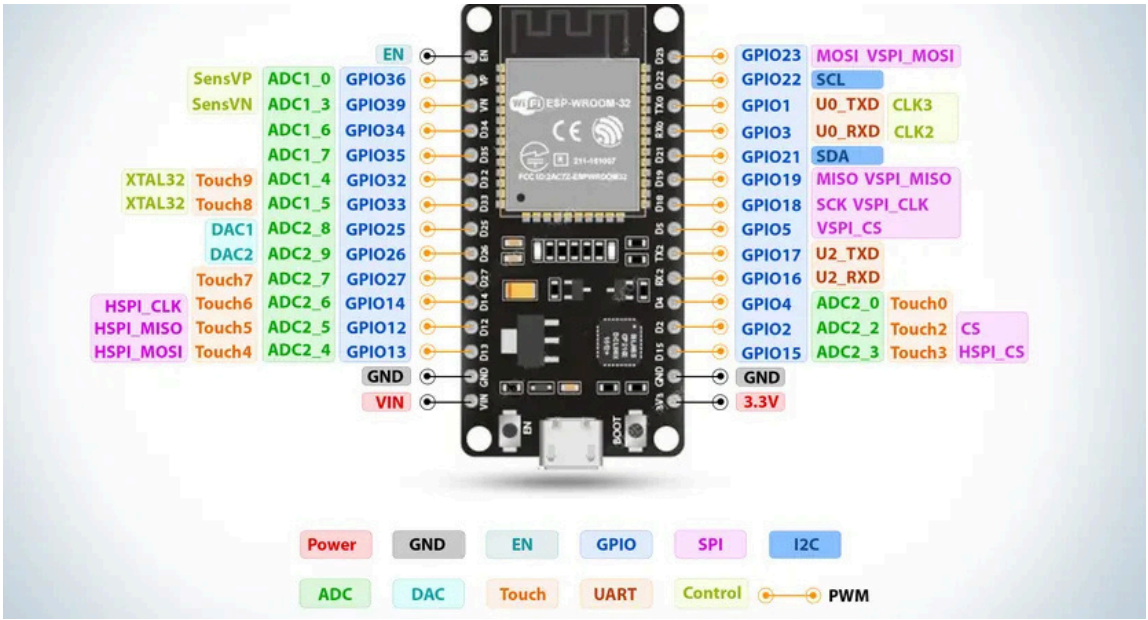


FIG 1: ESP 32 MODULE

FINGERPRINT SENSOR(R307) :-

The fingerprint sensor is used for voter authentication in the system. It scans and matches the user's fingerprint with stored data to verify identity. Since every fingerprint is unique, it ensures that only authorized users can vote and prevents duplicate voting. In this project, the sensor is connected to the ESP32 and communicates through serial communication. It provides a secure, fast, and reliable method of verification, making the voting system more accurate and trustworthy.

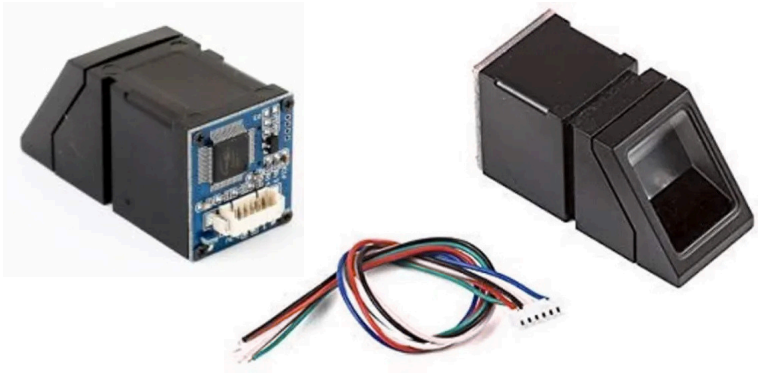


FIG 2: FINGERPRINT SENSOR

LCD DISPLAY 16X2

A 16 x 2 LCD Display is used to display the message about the votes , the candidates choose which party , it show's the name of all the party standing in the election's . In addition to this, it can be used to know what process is going on in the Microcontroller from Initialization of the devices. Its working voltage is 5V. It is interfaced with esp32 in 4-bit Operating Mode.

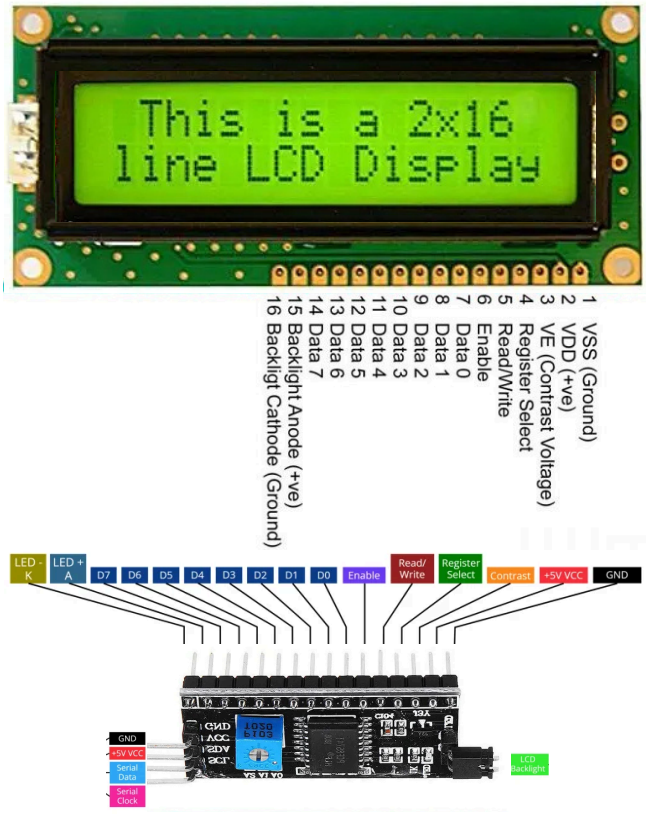


FIG 3: LCD DISPLAY

I2C MODULE

The I2C module is used to simplify communication between the ESP32 and the LCD. It reduces the number of wires by using only two pins (SDA and SCL). This makes the circuit simple and easy to connect while ensuring smooth data transfer.

PUSH BUTTON (MOMENTARY RESET)

Push buttons are used as input devices for candidate selection in the voting system. Each button is assigned to a specific candidate. They are connected to ESP32 GPIO pins and work by sending a signal when pressed. After successful fingerprint verification, the voter presses a button to cast their vote.



FIG 4: PUSH BUTTONS

BUCK CONVERTER

The buck converter is used to step down the input voltage to required levels such as 5V and 3.3V.

It provides a stable and efficient power supply to components like ESP32, LCD, and fingerprint sensor. It protects the system from voltage fluctuations and overheating. By maintaining proper voltage levels, it increases system safety and overall performance.

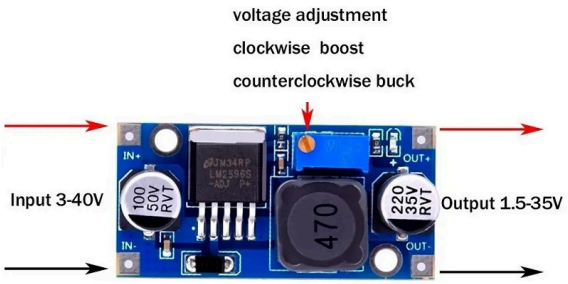


FIG 5: BUCK CONVERTER

RESISTOR'S

Resistors are used to control and limit the flow of current in the circuit. They help protect sensitive components from excess current and prevent damage. In this project, resistors are used with push buttons to ensure stable operation.

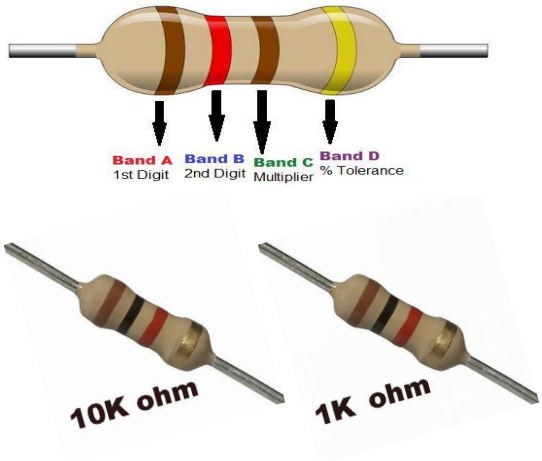


FIG 7: RESISTORS

BUZZER

It produces sound signals for actions like successful fingerprint verification, vote confirmation, or error alerts. This makes the system more user-friendly, as users can understand system responses without always looking at the display. It improves interaction and overall usability of the voting system.



FIG 6: BUZZER

BLOCK DIAGRAM :-

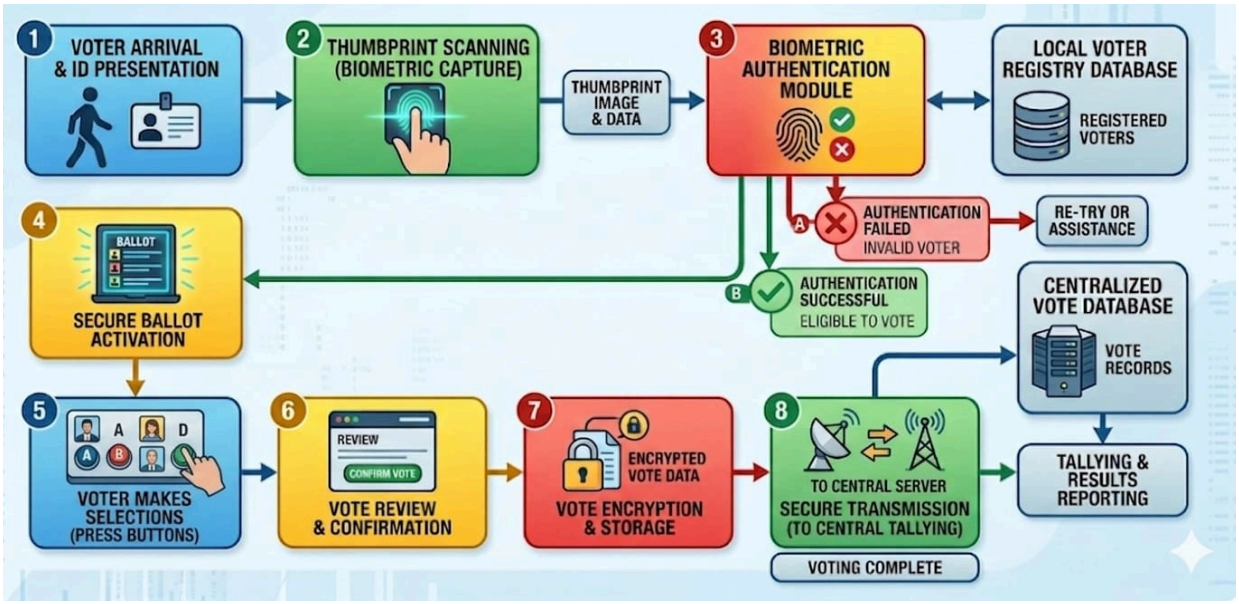


FIG 8: BLOCK DIAGRAM

CIRCUIT DIAGRAM :-

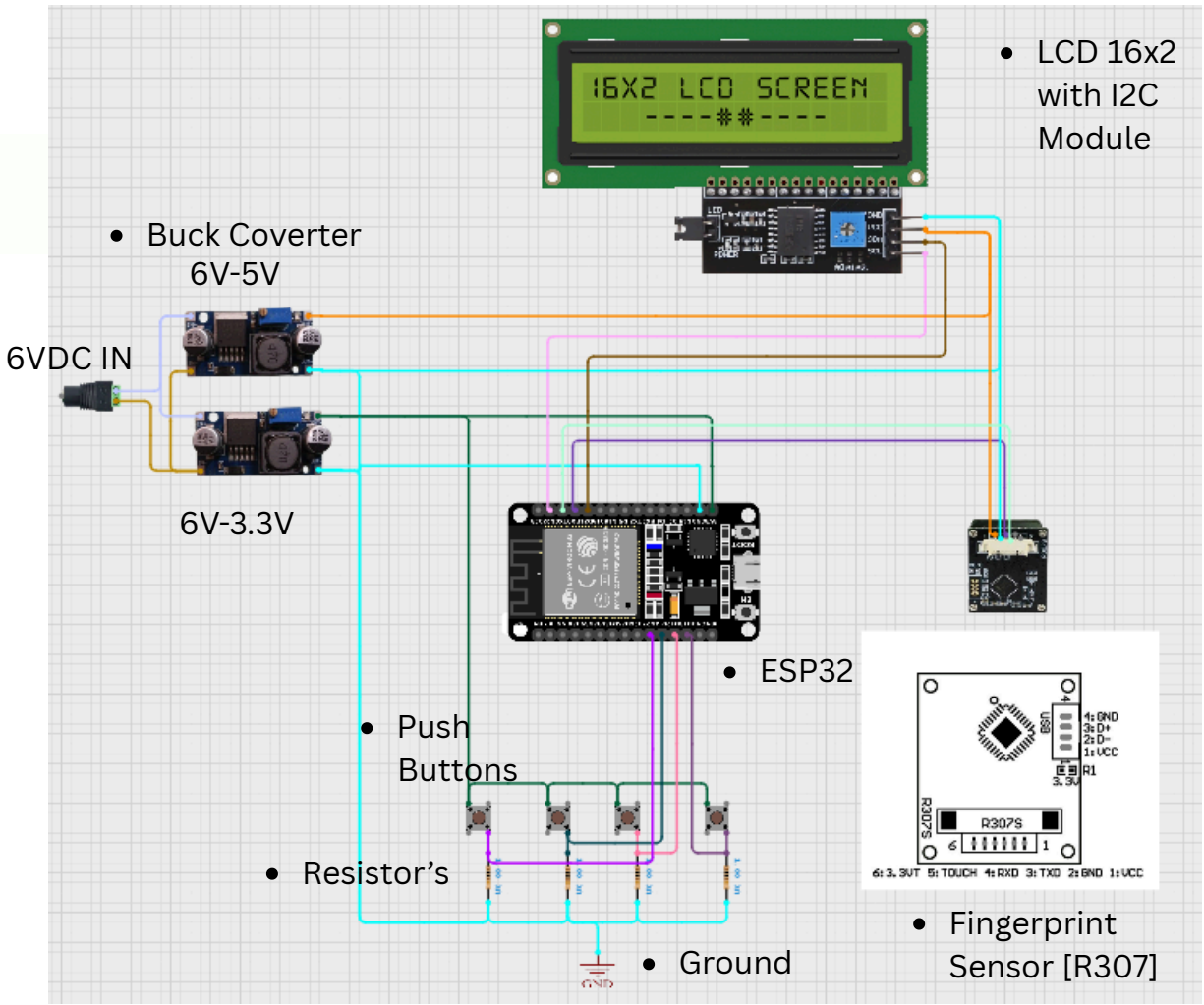


FIG 9: CIRCUIT DIAGRAM

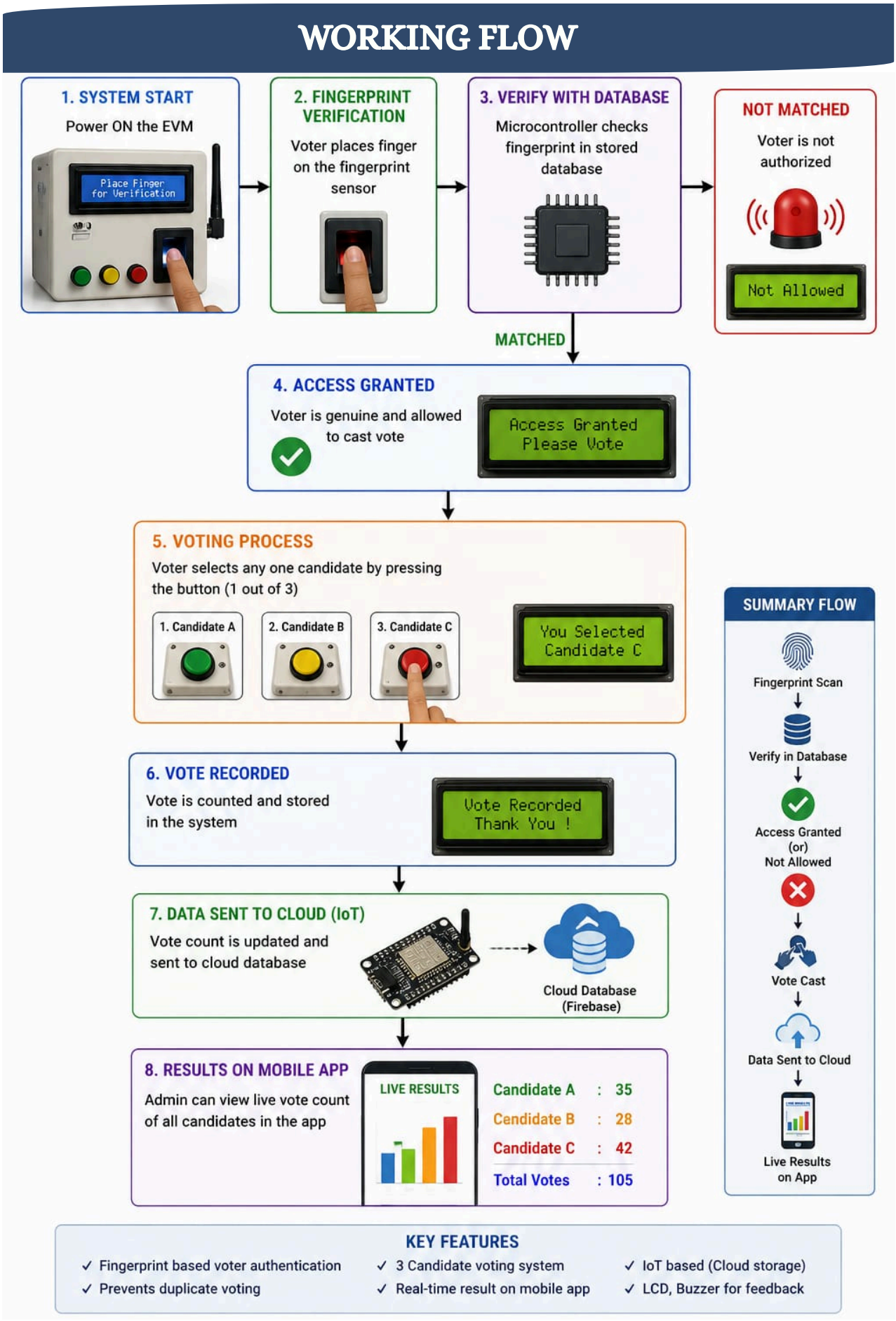


FIG 10: WORKING FLOW OF THE PROJECT

WORKING PRINCIPLE

- **Power Initialization**

The system is powered using a 6V DC adapter connected through a DC jack and the voltage is regulated using buck converters to provide required levels for different components while maintaining a common ground for proper operation.

- **System Startup**

When power is ON, the ESP32 initializes all connected peripherals, establishes Wi-Fi connection using stored credentials, and the LCD displays a welcome message such as "Place Finger" to guide the user**.**

- **Fingerprint Enrollment (Pre-setup) & Biometric Authentication**

Before voting begins, fingerprints of authorized users are enrolled and stored in the memory of the R307 sensor with unique IDs which are later used for identification**.**The voter places their finger on the sensor which captures the image, converts it into a digital template, and compares it with stored data to verify identity, allowing access only if a match is found.

- **Voting Process**

After successful authentication, the LCD displays candidate options and the voter selects a candidate using push buttons connected to ESP32.

- **Vote Recording**

ESP32 detects the button press through GPIO pins, records the vote by incrementing the respective candidate count, and ensures the voter cannot vote again using the same fingerprint ID.

- **Data Transmission (IoT)**

The recorded vote data is transmitted to an IoT platform using ESP32 Wi-Fi where it is updated and stored for real-time monitoring.

- **Result Display**

The LCD displays confirmation messages such as "Vote Recorded" and a buzzer may provide audio feedback while results are simultaneously updated on the web dashboard.

- **Duplicate Voting Prevention**

Once a fingerprint ID is used, it is marked as voted in the system which prevents the same user from voting again.

- **System Loop**

After completing one voting cycle, the system resets to its initial state and waits for the next user, continuously repeating the process.

ADVANTAGES

- **High security through biometric authentication**, ensuring only authorized users can vote and reducing fraud.
- **Prevents duplicate voting** by marking each verified voter ID after a vote is cast.
- **Accurate electronic vote counting, eliminating manual errors and improving reliability**.
- **Fast result processing** with automatic counting and instant updates.
- **Real-time IoT-based monitoring**, allowing results to be viewed remotely.
- **Transparent voting process** that builds trust among users and authorities.
- User-friendly **LCD** interface **that guides voters step by step**.
- **Reduced human intervention**, minimizing bias and manipulation.
- **Cost-effective implementation** using affordable and easily available components.
- **Wireless communication** using ESP32, removing the need for extra modules.
- **Compact and portable system design,** making it easy to deploy in different locations.
- **Low power consumption**, allowing efficient operation for long durations.
- **Easy maintenance and operation** compared to traditional voting systems.
- **Improved efficiency and reduced paperwork**, making the election process faster and more organized.
- **Quick voter verification process**, reducing waiting time during elections.
- **Useful for colleges, offices, societies, and organizations** where secure small-scale elections are required.

LIMITATIONS

- **Requires stable internet connection** for IoT features to function properly.
- **Limited fingerprint storage capacity** restricts the number of users.
- **Sensor accuracy affected by** finger condition such as **dirt or cuts**.
- **Dependency on power supply**, system fails if power is not available.
- **Initial enrollment required**, which takes time before actual voting
- **Hardware failure risk** in components like fingerprint sensor, LCD, or buttons can affect system operation.
- **Fingerprint recognition** can sometimes take extra time, causing delays during heavy usage.
- **Not suitable for very large-scale national elections** without further optimization and testing.
- **Data privacy concerns** may arise regarding the storage of biometric information.
- **Environmental conditions** such as dust, heat, or humidity may affect sensor performance.
- **System updates and software maintenance** are required regularly for smooth operation.

CONCLUSION

The **Advanced Electronic Voting Machine** , this project provides a secure, efficient, and modern solution to traditional voting problems. By integrating a fingerprint sensor with the ESP32 microcontroller, the system ensures that only authorized users can vote and eliminates the possibility of duplicate voting.

The use of IoT technology further enhances the system by enabling real-time result monitoring on a web platform, improving transparency and accessibility. The LCD interface makes the system easy to use, while automated vote counting reduces human effort and errors.

Overall, this project demonstrates how embedded systems and IoT can be combined to create a reliable and scalable voting solution. With further improvements, it can be adapted for larger and more complex election systems.

SCOPE OF FUTURE WORK

While the current prototype of the Advanced Electronic Voting Machine (EVM) successfully demonstrates secure biometric authentication and real-time voting confirmation, several advancements can be integrated to scale this system for large-scale, national elections.
The following methods could be adopted in future to scale the project further.

1) IoT Integration and Real-Time Centralized Counting
By utilizing the built-in Wi-Fi and Bluetooth capabilities of the ESP32 microcontroller, the system can be integrated with the Internet of Things (IoT).

• Cloud Database Sync: Instead of relying entirely on local memory, voter biometric data can be securely fetched from a centralized cloud database (such as the Aadhaar database in India).

• Live Result Compilation: Encrypted voting data can be transmitted in real-time to a secure central server, allowing election commissions to monitor voter turnout live and declare results within minutes of the polling close.

2) Multi-Factor Authentication (MFA)
To make the authentication process foolproof, the system can be upgraded from single-factor biometric (fingerprint) to multi-factor authentication

• RFID/Smart Card Reader: Voters must swipe a unique voter ID card embedded with an RFID chip.

• Facial Recognition: Incorporating a small camera module (like the ESP32-CAM) to perform facial recognition alongside fingerprint scanning to eliminate the possibility of spoofing.

3) Implementation of VVPAT (Voter Verifiable Paper Audit Trail)

maintain a physical backup of the digital votes, a micro thermal printer can be integrated into the system. When a voter presses a push-button, the printer will generate a small slip showing the candidate's name and serial number. The slip will be visible to the voter behind a glass window for a few seconds before automatically dropping into a sealed storage box.,

4) Advanced Power Management and Portability

For implementation in remote or rural areas with unstable power grids, future iterations can include:
Built-in Lithium-ion Battery Pack: Paired with a solar charging circuit to ensure uninterrupted operation.Deep Sleep Optimization: Programming the ESP32 to enter ultra-low power "deep sleep" mode between voters, waking up only when a new voter activates a primary switch.

5) Accessibility Features (Universal Design)
To make the EVM fully accessible to differently-abled citizens, the interface can be enhanced with:

- Braille Markings: Embossed Braille text alongside the tactile push-buttons for visually impaired voters.
- Voice-Guided Assistance: An audio jack or text-to-speech module connected to headphones, guiding the voter through the steps via voice prompts coordinated with the buzzer signals.

6) Unauthorised Access Detection Module (UADM) with Anti-Tamper Protocols
To protect the machine from physical tampering or unauthorized reverse-engineering during storage or transport, the system can be upgraded with an Unauthorised Access Detection Module (UADM).

• Physical Enclosure Sensors: The EVM casing would be equipped with continuous-monitoring micro-switches or light sensors. If anyone attempts to forcefully open the casing or drill into the PCB to access the ESP32 chip or memory lines the circuit will immediately detect a breach.
• Self-Destruct/Data-Wipe Routine: Upon detecting a physical breach, the microcontroller will trigger an automated security protocol that permanently wipes the encryption keys and voter logs from the internal memory, rendering the machine completely non-functional ("bricked").

• Real-Time Tamper Alerts: Since the ESP32 has wireless capabilities, it can simultaneously broadcast an automated emergency distress signal to a localized security mesh or central control room, timestamping the exact second the breach was attempted. This ensures complete custody-chain security from the manufacturing plant to the polling booth.

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- Project report templates and sample PDFs from academic websites for formatting guidance.